

Praveen Kaithepalli

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Summary

- AI & Robotics Engineer with 3+ years of experience delivering production-ready machine learning systems, robotic manipulation pipelines, and enterprise data solutions. Demonstrated success in reinforcement learning (PPO), deep imitation learning, NLP, 3D computer vision, simulation engineering, predictive modeling, data mining, and analytics, achieving measurable results such as 35% faster ETL pipelines, 25% improved stakeholder engagement, and 82% predictive accuracy in healthcare models. Skilled in modern C++, Python, PyTorch, software engineering, data management, management systems, construction, algorithms, disaster recovery, data acquisition, data structures, data modeling, data storage, Datastage, Snowflake, Hadoop, MySQL, NoSQL, MongoDB, DBT, and scalable architectures, with the ability to design and deploy robust systems that perform reliably in real-world environments.

Technical Skills

- **Programming:** Python, C++, Java, C#, R, Flex
- **AI & ML:** Deep Learning, Neural Networks, Computer Vision, NLP, Reinforcement Learning (PPO), Simulation, Algorithms
- **Libraries/Tools:** PyTorch, NumPy, Pandas, AutoGluon, Scikit-Learn
- **Data Engineering:** SQL, ETL, Data Warehousing, Big Data Tools (Hadoop ecosystem)
- **Visualization:** Tableau, Power BI
- **Cloud:** AWS, Google Cloud
- **Software Architecture:** API design, scalable data systems, performance-optimized pipelines
- **Other Domains:** Robotics fundamentals, Payroll systems, BI systems, Healthcare data, Legal services workflows
- **Soft Skills:** Communication, Collaboration, Problem-solving, Technical Documentation, Agile/Scrum

Volunteer Experience

Business Analyst (Volunteer), BrightMind, USA

Oct 2025 - Present

- Designed data flow models and KPIs for community-impact workflows.
- Built Power BI dashboards increasing donor engagement and transparency by 25%.
- Facilitated system design sessions to streamline analytics adoption.

Work Experience

Programmer Analyst - Cognizant Technology Solutions, India

July 2022 - July 2023

- Built & optimized ETL pipelines achieving 35% faster data refresh.
- Developed automated Python + SQL data quality checks increasing efficiency by 60%.
- Supported large-scale analytics, predictive modeling, and data warehousing architectures across healthcare & legal workflows.
- Applied NLP techniques to process unstructured healthcare and legal text data for improved reporting and analysis.
- Managed data acquisition, disaster recovery, and scalable storage solutions.

Programmer Analyst Trainee - Cognizant Technology Solutions, India

Jan 2022 - July 2022

- Designed and implemented ETL workflows using Snowflake, Hadoop, and Datastage.
- Authored BRDs, use cases, and data-flow diagrams capturing business logic.
- Applied data mining and NLP models to extract structured insights from unstructured datasets.
- Optimized SQL validation scripts achieving 20% faster execution.

Data Analyst Intern | AuroPro Soft Systems, India

Aug 2020 - Dec 2021

- Analyzed Oracle-based systems and constructed data models aligned with reporting needs.
- Automated reporting workflows reducing manual effort by 30%.
- Improved system stability and data storage reliability through root-cause analysis.
- Produced BRDs and process models for technical teams, supporting software engineering best practices.

Projects

- **Healthcare Process Optimization** - Confluence, JIRA, Lucidchart Aug 2025 - Sep 2025
Conducted end-to-end business analysis for a hospital management system using IBM tools; gathered requirements through stakeholder workshops and focus groups, created BRD and process flow diagrams, and performed fit-gap and root cause analyses. Improved workflow efficiency by 30% and enhanced reporting accuracy through optimized data modeling and SDLC documentation.
- **HR Analytics Dashboard - Power BI, SQL, Excel** Aug 2025 - Sep 2025
Built an interactive HR dashboard tracking KPIs (attrition 16.12%, average age 37, satisfaction 4.1/5); automated data refresh with Power Query, improving reporting efficiency by 35% and data accuracy by 30%.
- **Healthcare Predictive Analytics - NHANES Dataset** Jan 2025 - May 2025
Applied data modeling and statistical analysis to predict patient outcomes with 82% accuracy. Designed executive Tableau dashboards that reduced data interpretation time by 15%.

- Loan Approval Optimization (R, ML)**

Aug 2024 - Dec 2024

 Built machine learning models (XGBoost, Logistic Regression) automating approvals with 81% accuracy. Developed Power BI dashboards to visualize loan trends, risk levels, and compliance metrics.
- Robotic Learning & Manipulation**
 Developed RL (PPO) and imitation learning agents for autonomous decision-making. Implemented vision-based pipelines for grasping, 3D perception, and environment interaction. Designed simulation tools for manipulation tasks including re-grasping & pick-and-place. Modeled uncertainty from sensors and physical interactions for robust behavior.
- 3D Perception Pipeline**
 Implemented depth-based perception, point cloud processing, and 3D reconstruction for object pose estimation.
- NLP Projects**

Aug 2024 - Dec 2024

 Developed transformer-based models for text embedding and classification tasks. Applied sequence modeling to extract structured insights from unstructured healthcare and legal documents. Optimized NLP pipelines for production-level integration with analytics dashboards.

Certifications

IBM Business Analyst	Sep 2025 - Present
IBM Capstone Project: Applying Business Analysis Skills	Aug 2025 - Sep 2025
SQL for Data Science	Aug 2025 - Sep 2025
SAS Programming - Getting Started with SAS	July 2025 - Aug 2025
AWS - Machine Learning Foundations	May 2020 - June 2020

Education

University at Albany, SUNY	Aug 2023 - May 2025
Master of Science in Data Science/ Applied Mathematics	Albany , NY
Relevant Coursework: Machine Learning • Statistics & Probability • Statistical Analysis • Data Visualization • Cloud Computing	